

શ્રી રાજીવભાઈ ભટ્ટ

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અંગરેજી લે (Book) લેખક અને જાણી
શરૂ કરી અને તેમાં પ્રવેશ કરાવે
શરૂ કરી અને તેમાં પ્રવેશ કરાવે

સંસ્કૃત

Box "Mathematics is my life."

→ Linear Transformation

→ Real Analysis:

(iii) If T_1 and T_2 are two linear transformations from $U(F)$ to $V(F)$

$$\text{i.e. } T_1, T_2: U(F) \rightarrow V(F)$$

then $(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha)$ is again a Linear Transformation $\forall \alpha \in U$ from $U(F)$ to $V(F)$.

$$\text{And also } (CT_1)(\alpha) = CT_1(\alpha) \\ \forall \alpha \in U \text{ \& } C \in F$$

is again a L.T from $U(F)$ to $V(F)$.

$$\begin{aligned} \text{Proof: } (T_1 + T_2)(\alpha + \beta) &= T_1(\alpha + \beta) + T_2(\alpha + \beta) \\ &= \alpha T_1(\alpha) + T_1(\beta) \\ &\quad + \alpha T_2(\alpha) + T_2(\beta) \\ &= (T_1 + T_2)(\alpha) + (T_1 + T_2)(\beta) \\ &= \alpha (T_1 + T_2)(\alpha) + (T_1 + T_2)(\beta) \end{aligned}$$

$$\therefore (T_1 + T_2)(\alpha + \beta) = \alpha (T_1 + T_2)(\alpha) + (T_1 + T_2)(\beta)$$

Next,

$$(CT_1)(\alpha)$$

$$\begin{aligned} (CT_1)(\alpha + \beta) &= C T_1(\alpha + \beta) \\ &= C [\alpha T_1(\alpha) + T_1(\beta)] \end{aligned}$$

$$= a (CT_1)(\alpha) + (CT_1)(\beta)$$

$$\therefore (CT_1)(\alpha + \beta) = a (CT_1)(\alpha) + (CT_1)(\beta).$$

Corollary: $T: V(F) \rightarrow V(F)$ be L.T then

$\sum c_i T_i$ is again a L.T : iff

Ques) Consider transformation $T: P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R})$ given by

$$T(P(x)) = 7P'(x) + 11 \int_0^x P(x) dx.$$

Find Rank and Nullity of T ?

Sol:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

$$p'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1}$$

$$\therefore T(a_0 + a_1x + a_2x^2 + \dots + a_nx^n)$$

$$= 7(a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1})$$

$$+ 11 \int_0^x (a_0 + a_1x + \dots + a_nx^n) dx$$

$$= 7a_1 + 7 \cdot 2 a_2 x + 7 \cdot 3 a_3 x^2 + \dots + 7 \cdot n a_n x^{n-1}$$

$$+ 11 a_0 x + \frac{11}{2} a_1 x^2 + \frac{11}{3} a_2 x^3 + \dots + \frac{11}{n+1} a_n x^{n+1}$$

$$+ \frac{11}{n} a_{n-1} x^n + \frac{11}{n+1} a_n x^{n+1} = 0$$

$$\text{iff } a_n = 0 \Rightarrow a_n = a_{n-2} = \dots = 0.$$

$$\text{and } a_{n-1} \neq 0 \Rightarrow a_{n-1} = a_{n-3} = \dots = 0.$$

$$\therefore T(p(x)) = 0 \Rightarrow p(x) = 0 \text{ only}$$

$$\therefore N(T) = \{0\}$$

$$\therefore \begin{cases} \text{Nullity}(T) = 0 \\ \text{Rank}(T) = n+1 \end{cases} \quad \begin{array}{l} \therefore \text{This is the Ex of} \\ \text{non-singular trans.} \\ \text{as } \dim p_n(V) = n+1 \end{array}$$

* Singular & Non-singular Transformations

$\hookrightarrow$ A transformation $T: V(F) \rightarrow V(F)$ is said to be Non-singular Transformation if null space of the transformation is zero space

$$\text{i.e. Nullity}(T) = 0.$$

if $\text{Nullity}(T) \geq 1$, then transformation T is called singular.

$\rightarrow$ Defⁿ: A Transformation $T: V(F) \rightarrow V(F)$ is said to be one-one (or) one to one or Injective if,

$$T(a_1) = T(a_2) \Rightarrow a_1 = a_2$$

$$\forall a_1, a_2 \in V(F)$$

$$\text{or } a_1 \neq a_2 \Rightarrow T(a_1) \neq T(a_2) \quad \forall a_1, a_2 \in V(F)$$

Note: $T(d_1) - T(d_2) = \hat{0}$

$$\Rightarrow T(d_1 - d_2) = \hat{0}$$

$$\Rightarrow (d_1 - d_2) \in N(T)$$

$$\therefore d_1 - d_2 = \sum_{i=1}^n c_i d'_i$$

where,

$$B_{N(T)} = \{d'_1, d'_2, \dots, d'_n\}$$

$$\therefore d_1 = d_2 + \sum c_i d'_i$$

$$\Rightarrow d_1 = d_2 \text{ only if } N(T) \text{ is zero space.}$$

$\hookrightarrow$ A Transformation $T: U(F) \rightarrow V(F)$ is one-one or injective iff $\text{Nullity}(T) = 0$.

i.e. $\text{Nullity}(T) > 0$

i.e. $\Rightarrow T$ is not one-one

i.e. $\Rightarrow T$ is many-one transformation

Note: $\text{Nullity}(T) = 0$ for one-one transformation

$$\Rightarrow \boxed{\text{Rank}(T) = \dim U}$$

and $\text{rank}(T) \leq \dim V$ as range of T is s. space of V .

$$\Rightarrow \boxed{\dim U \leq \dim V}$$

④ A necessary condition for a transformation $T: U(F) \rightarrow V(F)$ to be one-one is that

$$\boxed{\dim U \leq \dim V}$$

Def. A function whose $\boxed{\text{Range} = \text{codomain}}$ is said to be **Onto function**. It is also known as **Surjective**.
and if $\boxed{\text{Range} \subset \text{codomain}}$ then function is known as **Into function** (Not onto)

$T: U(F) \rightarrow V(F)$ will be onto if

$$\boxed{\text{Rank}(T) = V}$$

$$\Rightarrow \text{Rank}(T) = \dim V$$

But $\text{Rank}(T) \leq \dim U$ as $\text{Rank}(T) \leq \dim U$

$$\Rightarrow \boxed{\dim V \leq \dim U}$$

⑤ A necessary condition for a onto transformation $T: U(F) \rightarrow V(F)$ to be is that

$$\boxed{\dim U \geq \dim V}$$

Defⁿ:- A Transformation $T: U(F) \rightarrow V(F)$ is
= bijective or one-one onto transformation
if

1) T is one-one $\Rightarrow \dim U \leq \dim V$

2) T is onto, $\Rightarrow \dim U \geq \dim V$

$\Rightarrow$ A necessary and condition for a
Transformation $T: U(F) \rightarrow V(F)$ to be
bijective or one-one onto or
Invertible is that

$$\boxed{\dim U = \dim V}$$

Note:- If $\dim U = \dim V$ & $\text{Nullity}(T) = 0$

$\Rightarrow \text{Rank}(T) = \dim U = \dim V$

one-one $\Rightarrow$ onto.

if onto $\Rightarrow \text{Rank}(T) = \dim V$

* $\text{Rank}(T) = \dim U \Rightarrow \text{Nullity}(T) = 0$.

$\Rightarrow T$ is one-one

$\Rightarrow$ if T is onto $\Rightarrow T$ is one-one

If U & V are two vector space
over same field with same dim
then -

(1) if T is one-one $\Rightarrow T$ is onto

(2) if T is onto $\Rightarrow T$ is one-one

i.e. Either it is both one-one &
onto

(or)

neither one-one, nor onto

Defⁿ

(Defⁿ) If T is Bijection, $T: U(F) \rightarrow V(F)$
then a function $T^{-1}: V(F) \rightarrow U(F)$
is said to be Inverse of T
if, $T^{-1}(\beta) = \alpha$ whenever $T(\alpha) = \beta$
 $\forall \alpha \in U, \beta \in V$

(Ques) A Linear Transformation

$$T: \mathbb{R}^{10}(\mathbb{R}) \rightarrow \mathbb{R}^6(\mathbb{R})$$

1) may be one-one

2) may be onto

3) may not be one-one

4) may not be onto.

Solⁿ

$$\text{Rank}(T) \leq 6$$

$$\Rightarrow \text{nullity}(T) \geq 4$$

$\therefore$ Null space have atleast 4 basis
weighting constant

Ques) consider Transformation

$T: P_2(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ then which of
the following

- 1) may be one-one
- 2) may be onto
- 3) may not be one-one
- 4) may not be onto.

Solⁿ

$$\dim V = 3$$

$$\dim W = 3$$

$$\text{Rank}(T) \leq 3$$

When $\text{Rank}(T) = 3 \Rightarrow \text{nullity}(T) = 0$

$\Rightarrow T$ may be one-one

$$(2) \text{Rank}(T) \leq 3 \neq \dim(\mathbb{R}^3(\mathbb{R}))$$

(1) If $\text{rank}(T) = 4$ then $\text{Nullity}(T) = 2$
 $\therefore T$ may not be one-one

(2) consider $T: P_5(P) \rightarrow M_{22}(R)$

~~(1)~~ may be one-one

~~(2)~~ may be onto

~~(3)~~ may be not be one-one

~~(4)~~ may not be onto.

~~(5)~~ may be both one-one & onto.

~~(6)~~ may be neither one-one & onto.

~~(7)~~ may be one-one but not onto.

~~(8)~~ may be onto but not one-one

soln: $\dim V = 6 = \dim W$

$$\text{rank}(T) \leq 6$$

(1) If $\text{rank}(T) = 6 \Rightarrow \text{Nullity}(T) = 0$
 $\dim V$

$\therefore T$ is onto.

(2) If $\text{rank}(T) = 6 \Rightarrow \text{Nullity}(T) = 0$

$\therefore T$ is one-one

(3) If $\text{rank}(T) = 5 \Rightarrow \text{Nullity}(T) = 1$

$\therefore T$ is not one-one

If $\text{rank}(T) = 4 \Rightarrow \text{Nullity}(T) = 2$
 $\therefore T$ is not onto.

Ques) consider vector space V of all twice differentiable functions over field of real numbers satisfying

$$f'' - 2f' + f = 0 \text{ and further}$$

Transformation $T: V \rightarrow \mathbb{R}^2(\mathbb{R})$ given by

$$T(f(x)) = (f'(0), f(0)) \text{ then}$$

1) T is one to one & onto.

2) T is one one but not onto

3) T is onto but not 1-1

4) T is neither 1-1 nor onto.

In the above question, T is invertible then also find $T^{-1} = ?$

Solⁿ

$$f'' - 2f' + f = 0.$$

$$\Rightarrow (D^2 - 2D + 1)f = 0.$$

$$\begin{aligned} \Rightarrow A.E &= m^2 - 2m + 1 \\ &= (m-1)(m-1) \\ &= (m-1)^2 = 0. \end{aligned}$$

$$f(x) = c_1 e^x + c_2 x e^x$$

$$V(m) = \left\{ c_1 e^x + c_2 x e^x \mid c_1, c_2 \in \mathbb{R} \right\}$$

the given transformation is

$$T(f_1 e^x + f_2 x e^x) = (f_1 + f_2, f_1)$$

$$f(0) \rightarrow f(0) = f_1$$

$$\begin{aligned} f'(0) &= f_1 e^0 + f_2 e^0 + f_2 x e^0 \\ &= f_1 e^0 + f_2 (e^0 + x e^0) \end{aligned}$$

$$f'(0) = f_1 + f_2$$

$$\therefore T(f_1 e^x + f_2 x e^x) = (0) \iff (f_1 + f_2, f_1) = (0, 0)$$

$$\Rightarrow f_1 + f_2 = 0 \text{ \& } f_1 = 0$$

$$\Rightarrow f_2 = 0 \text{ \& } f_1 = 0.$$

$$\text{Nul}(T) = \{0\}$$

$$\therefore \text{Nullity}(T) = 0$$

$\therefore T$ is one-one

and basis of $V(W)$ is

$$B_V = \{e^x, x e^x\} \quad ?$$

$$\therefore \dim V(W) = 2$$

a. By Rank-Nullity theorem

$$0 + \text{Rank}(T) = \dim V$$

$$\text{Rank}(T) = 2$$

$$\therefore \text{Range}(T) = 2 = \dim \mathbb{R}^2(\mathbb{R})$$

$\Rightarrow T$ is one-one and onto.

Next if T is L.T. s.t.

$$T(c_1 e^x + c_2 x e^x) = (c_1 + c_2, c_1)$$

$$\Rightarrow T^{-1}: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathcal{V}(\mathbb{R})$$

$$T^{-1}(c_1 + c_2, c_1) = c_1 e^x + c_2 x e^x$$

$$\text{Let } (c_1 + c_2, c_1) = (a, b)$$

$$\Rightarrow \text{not } \emptyset. \quad \Rightarrow c_1 + c_2 = a$$

$$\Rightarrow c_1 = b \Rightarrow c_2 = a - b.$$

$$\therefore T^{-1}(a, b) = b e^x + (a - b) x e^x$$

$$= b e^x + a x e^x - b x e^x$$

$$= a x e^x + b (e^x - x e^x)$$

$$\therefore T^{-1}(a, b) = a x e^x + b (e^x - x e^x)$$

(Q4c) Let T is L.T, $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$
given by

$$T(a, b, c) = (a + b, b + c, c + a)$$

Is T invertible Transformation if yes
show find T^{-1} .

$$T(a, b, c) = (0, 0, 0)$$

$$\Rightarrow (a+b, b+c, c+a) = (0, 0, 0)$$

$$\Rightarrow \begin{cases} a+b=0 \\ b+c=0 \\ c+a=0 \\ \hline a+b+c=0 \end{cases}$$

$$\Rightarrow a=0, b=0, c=0.$$

$$\therefore \text{Nullity}(T) = 0.$$

$\therefore T$ is one-one.

$\therefore$ Then T is onto also. as $\dim V = \dim W$.

$$\text{Now } T^{-1}: \mathbb{R}^3(W) \rightarrow \mathbb{R}^3(V)$$

$$T^{-1}(a+b, b+c, c+a) = (a, b, c)$$

$$\text{Let } a+b = x$$

$$b+c = y$$

$$c+a = z$$

$$a(a+b+c) = x+y+z$$

$$a+b+c = \frac{x+y+z}{2}$$

$$\Rightarrow CQ = \frac{y+z-x}{2}$$

$$aQ = \frac{x+z-y}{2}$$

$$bQ = \frac{x+y-z}{2}$$

$$\text{Hence } T^{-1}(x, y, z) = \left\{ \frac{x+y+z}{2}, \frac{x-y+z}{2}, \frac{x+y-z}{2} \right\}$$

(Case) if T is a linear transformation, $T: V(F) \rightarrow V(F)$ where $\dim V(F) = n$.

then which of the following options is sufficient / guarantees / ensures / necessitates implies that T is bijective.

1) $\text{Rank}(T) + \text{Nullity}(T) = n$

2) $\text{Rank}(T) - \text{Nullity}(T) = n$

3) $\text{Rank}(T) + 3 \text{Nullity}(T) = n$

4) $2 \text{Rank}(T) + 7 \text{Nullity}(T) = 2n$

5) $2 \text{Rank}(T) + \text{Nullity}(T) = n$

6) Rank Nullity theorem holds for all transformations.

$\therefore$ given implies that T is bijective.

we have

$$\textcircled{1} \quad f(T) + v(T) = 0 \cdot n \rightarrow \textcircled{1}$$

$$\text{suppose } f(T) - v(T) = n \rightarrow \textcircled{2}$$

$$\Rightarrow f(T) = n, \quad v(T) = 0$$

$\therefore T$ is one-one & onto.

$\therefore T$ is bijective

$$\textcircled{2} \quad f(T) + v(T) = n \rightarrow \textcircled{1}$$

$$f(T) + 3v(T) = n \rightarrow \textcircled{2}$$

$$\Rightarrow f(T) = n, \quad v(T) = 0$$

$\Rightarrow T$ is one-one & onto

$\therefore T$ is bijective.

$$4) \quad f(T) + v(T) = n.$$

$$2f(T) + 7v(T) = 2n$$

$$\Rightarrow f(T) = n \quad \& \quad v(T) = 0.$$

$\therefore T$ is one-one & onto.

$\therefore T$ is bijective

⑤

$$S(T) + r(\eta) = \eta$$

$$S(T) + r(\eta) = \eta$$

$$S(T) = 0 \rightarrow r(\eta) = \eta$$

$\therefore T$ is not one-one

$\therefore T$ is not bijective.

Que)

Let S be linear transformation

$S: V(F) \rightarrow W(F)$ & T is another

transformation $T: U(F) \rightarrow V(F)$.

Then product of linear transformation

is denoted by $ST: U(F) \rightarrow V(F)$.

defined by

$$(ST)(\alpha) = S(T(\alpha))$$

The above is also known as product
or composition of linear transformations.

Proof: $\forall \alpha \in F$ & $d_1, d_2 \in V$

$$ST(\alpha d_1 + d_2) = S[T(\alpha d_1 + d_2)]$$

$$= S[\alpha T(d_1) + T(d_2)]$$

$\because T$ is L.T

$$= \alpha(ST)(d_1) + (ST)(d_2)$$

$\therefore ST$ is a L.T

if T is linear operator on $V(F)$

i.e. $T: V(F) \rightarrow V(F)$ then

$$T^0 = I: V(F) \rightarrow V(F)$$

$$T^k = \underbrace{T \circ T \circ \dots \circ T}_{k \text{ times}}: V(F) \rightarrow V(F)$$

(ex) $T_1: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ given by

$$T_1(x, y) = (y, x)$$

$T_2: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ given by

$$T_2(x, y) = (0, y)$$

$T_1 T_2: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ is $\neq T_2 T_1: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$

folw:

$$T_1 T_2(0, y) = T_1(T_2(0, y))$$

$$= T_1(0, y)$$

$$= T_1(0, y)$$

$$= (y, 0)$$

$$T_2 T_1(x, y) = T_2(T_1(x, y))$$

$$= T_2(y, x)$$

$$= (0, x)$$

$$\Rightarrow T_1 T_2 \neq T_2 T_1$$

In general, if T_1 & T_2 are L.T then $T_1 \circ T_2$ may or may not be equal to $T_2 \circ T_1$

28/9/2015

morning

Ques: consider vector space of all real polynomial over field of real numbers and transformation given by

$$S: V \rightarrow V \quad \begin{matrix} ST(u) = S[T(u)] \\ \text{given by} \end{matrix}$$

$$S(2ax^i) = a_i x^{i+1} \\ = S(a_0 + a_1 x + \dots + a_n x^n)$$

$$T(2ax^i) = D(ax^i) \quad ; \quad D = \frac{d}{dx}$$

find ST & TS and discuss whether they are equal or not?

for

$$ST(a_0 + a_1 x + a_2 x^2 + \dots)$$

$$= S(a_1 + 2a_2 x + 3a_3 x^2 + \dots)$$

$$= (a_1 x + 2a_2 x^2 + 3a_3 x^3 + \dots)$$

$$TS(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots)$$

$$= T(a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + \dots)$$

$$= (a_0 + 2a_1 x + 3a_2 x^2 + \dots)$$

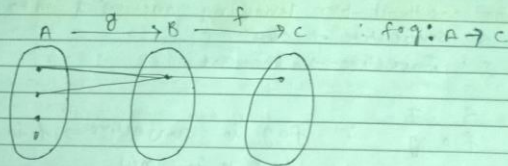
Now $(TS - ST) = I$

$(TS - ST)(a_0 + a_1x + \dots)$

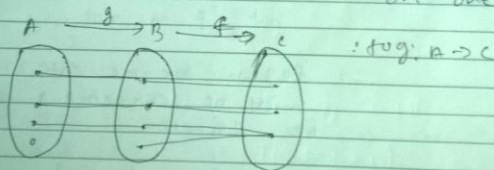
$= a_0 + a_1x + a_2x^2 + \dots$

$= I$

$\therefore ST \neq TS$



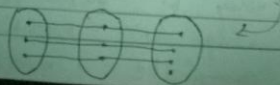
If g is not one-one $\Rightarrow f \circ g$ is not one-one.



If $f \circ g$ is one-one $\Rightarrow g$ is one-one

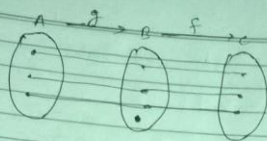
but If $f \circ g$ is one-one then do not implies that f is one-one.

$\hookrightarrow f \circ g$ is onto $\Rightarrow f$ is onto.



S: surjective
I: injective

S I
f o g:



f o g is onto that does not implies that g is onto.

⇒ Short method for knowing one-one & onto

↳ S: surjective → onto

I: injective → one-one

S I

f o g

∴ f o g is surjective ⇒ f is surjective but g is not

∴ f o g is injective ⇒ g is injective but f is not.

∴ f o g is one-one onto ⇒ f must be surjective & g must be injective

Ques/ Let $S: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ and $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ then which of the following are true

✓ 1) S o T may be one-one

✗ 2) S o T may be onto

✓ 3) T o S may be one-one

✓ 4) T o S may be one-one onto.

✗ 5) S o T may be invertible

✓ 6) T o S may be invertible

S may be one-one, not onto.
T may be onto but not one-one

Let: $S: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ & $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$
 $\hookrightarrow (1)$ $\hookrightarrow (2)$

In (1) $\dim U = \dim V$ also in (2) $\dim U = \dim V$

$\therefore S$ is one-one & onto
 $S: \mathbb{R}^3 \rightarrow \mathbb{R}^4$: $\dim U < \dim V$: one
 S is one-one but not onto
 $S \circ T \rightarrow$ is one-one as
 $\therefore T$ is onto but not one-one & S is one-one but not onto

$\hookrightarrow S \circ T$ is one-one iff T is one-one but T is not one-one

$\hookrightarrow S \circ T$ is onto iff S is onto but S is not onto
 $\therefore S \circ T$ is not invertible (check it)

S I
 T O S

$\hookrightarrow T \circ S$ is one-one iff S is one-one and here S is one-one
 $\therefore T \circ S$ is one-one

$\hookrightarrow T \circ S$ is onto iff T is onto and here T is onto

$\therefore T \circ S$ is onto

$\therefore T \circ S$ is invertible (check it)

Ques) If v_1, v_2, v_3, v_4 are non-zero vectors

$$\phi_1: v_1 \rightarrow v_2; \quad \phi_2: v_2 \rightarrow v_3;$$

$$\phi_3: v_3 \rightarrow v_4; \quad \text{such that}$$

$$\ker \phi_1 = \text{zero space} = \{0\}$$

$$\text{Range}(\phi_1) = \ker(\phi_2); \quad \text{Range}(\phi_2) = \ker(\phi_3)$$

$$\text{Range}(\phi_3) = \ker(\phi_4) \quad \forall \phi_4 \quad \text{then}$$

which of the following are true.

$$\checkmark (1) \sum_{i=1}^4 (-1)^i \dim v_i = 0.$$

$$\checkmark (2) \sum_{i=2}^4 (-1)^i \dim v_i > 0.$$

$$\times (3) \sum_{i=1}^4 (-1)^i \dim v_i < 0.$$

$$\times (4) \sum_{i=1}^4 (-1)^i \dim v_i \neq 0.$$

Solⁿ $\ker \phi_1 = \{0\}$

$$\Rightarrow N(1) = 0.$$

$$\therefore \text{Rank}(\phi_1) = \dim v_1 = \text{Nullity } \phi_2$$

$$(\because \text{Range}(\phi_1) = \ker(\phi_2))$$

$$\Rightarrow \dim \phi_1 = \dim \phi_2$$

$$\dim v_1 = \text{Nullity } \phi_2$$

$$\text{Range } \phi_2 = \text{Ker}(\phi_3)$$

$$\Rightarrow \text{Rank } \phi_2 = \text{Nullity}(\phi_3)$$

$$\text{Rank } \phi_2 = \dim v_2 - \dim v_1 = \text{Nullity}(\phi_3)$$

$$\text{Rank}(\phi_3) = \dim v_3 - (\dim v_2 - \dim v_1) = \dim v_4$$

$$\therefore \dim v_3 - \dim v_2 + \dim v_1 = \dim v_4$$

$$\therefore -\dim v_1 + \dim v_2 - \dim v_3 + \dim v_4 = 0$$

$$\therefore \sum_{i=1}^4 (-1)^i \dim v_i = 0$$

$$\text{also, } -\dim v_2 + \dim v_3 = \dim v_4$$

$$\therefore \dim v_4 - \dim v_3 + \dim v_2 > 0 \quad \text{or} \quad \text{dim nonzero space is } \geq 1$$

$$\therefore \sum_{i=2}^4 (-1)^i \dim v_i > 0$$

$$\boxed{\text{Rank}(AB) \geq \text{Rank}(A) + \text{Rank}(B) - n}$$

(Ques) If T is a square matrix of order 4 such that $T^4 = 0$. Let us denote

$k_i = \dim \text{Ker}(T^i)$, then which of the followings an inequality for

$k_1 \leq k_2 \leq k_3 \leq k_4$ will not be possible.

$$X1) 1 \leq 2 \leq 3 \leq 4$$

$$L2) 1 \leq 3 \leq 4 \leq 4$$

$$L3) 1 \leq 4 \leq 4 \leq 4$$

$$L4) 1 \leq 2 \leq 4 \leq 4$$

for i

$$K_i = \dim(\ker(T^i))$$

$$= \text{Nullity}(T^i)$$

$$\therefore \text{Nullity}(T) = 1$$

$$\begin{aligned} \therefore \text{Rank}(T) &= \text{order}(A) - \text{Nullity} \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \therefore \text{Rank}(T^2) &\geq \text{Rank}(T) + \text{Rank}(T) - n \\ &\geq 3 + 3 - 4 = 2 \end{aligned}$$

$$\therefore \text{Rank}(T^2) \geq 2$$

$$\therefore \text{Nullity}(T^2) = 4 - (\geq 2) \leq 2$$

$$\begin{aligned} \therefore \text{Rank}(T^3) &\geq \text{Rank}(T^2) + \text{Rank}(T) - 4 \\ &\geq 2 + 3 - 4 \\ &\geq 1 \end{aligned}$$

$$\therefore \text{Nullity}(T^3) \leq 3$$

$$\text{Rank}(T^4) \geq \text{Rank}(T^3) + \text{Rank}(T) - 4$$

$$\geq 3 + 1 + 3 - 4$$

$$= 0.$$

$$\therefore \text{Nullity}(T^4) \leq 4$$

$\therefore 1 \leq 2 \leq 3 \leq 4$ which is possible then remaining are not possible, so answer is (3), (6) & (4). //

* L.T as Vectors.

* $\text{Hom}(U, V)$ or $L(U, V)$

↳ If $U(F)$ & $V(F)$ are two vector spaces over same field F , then set of all L.T from U to V form a vector space, also denoted by

$$L(U, V) = \text{Hom}(U, V) \text{ w.r.t vector}$$

$$\text{addition as } (T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) ; \forall \alpha \in U$$

and scalar multiplication as

$$(cT)(\alpha) = cT(\alpha) ; \forall \alpha \in U.$$

$$\text{↳ } \left\{ \begin{array}{l} \text{If } \dim U = m \text{ \& } \dim V = n, \text{ then} \\ \dim(L(U, V)) = mn \end{array} \right\}$$

Proof: or working rule.

$$\text{Let } B_U = \{\alpha_1, \alpha_2, \dots, \alpha_m\} \text{ \& }$$

$$B_V = \{\beta_1, \beta_2, \dots, \beta_n\} \text{ be ordered basis of } U \text{ \& } V \text{ respectively.}$$

then let us define a linear transformation
 T_{ij} which is such that,

$$T_{ij}(d_j) = \beta_j \quad \&$$

$$T_{ij}(d_i) = \hat{0}$$

$$\text{i.e. } T_{11}(d_1) = \beta_1$$

$$T_{11}(d_2) = \hat{0}$$

$$T_{11}(d_3) = \hat{0}$$

$$\vdots$$

$$T_{1n}(d_n) = \hat{0}$$

$$T_{23}(d_1) = \hat{0}$$

$$T_{23}(d_2) = \beta_3$$

$$T_{24}(d_3) = \hat{0}$$

Now, let

$$B = \{T_{ij} \mid 1 \leq i \leq n, 1 \leq j \leq n\}$$

$\hookrightarrow$ claim: we have to prove that B is L.I.
 a $L(B) = L(V \otimes V)$.

(1) B is L.I.

$$c_{11}T_{11} + c_{12}T_{12} + \dots + c_{1n}T_{1n}$$

$$+ c_{21}T_{21} + c_{22}T_{22} + \dots + c_{2n}T_{2n}$$

$$\vdots$$

$$+ c_{n1}T_{n1} + c_{n2}T_{n2} + \dots + c_{nn}T_{nn} = \text{zero}$$

Transformation

$$\Rightarrow (a_1 d_1 + a_2 d_2 + \dots + a_n d_n) = 0$$

w/ $a_1, a_2, \dots, a_n \in F$

$$\Rightarrow a_1 (c_{11} \beta_1 + c_{12} \beta_2 + \dots + c_{1n} \beta_n) \\ + a_2 (c_{21} \beta_1 + c_{22} \beta_2 + \dots + c_{2n} \beta_n)$$

...

$$+ a_m (c_{m1} \beta_1 + c_{m2} \beta_2 + \dots + c_{mn} \beta_n) = 0 \text{ is true}$$

w/ $a_1, a_2, \dots, a_m$ only if

$$c_{11} \beta_1 + c_{12} \beta_2 + \dots + c_{1n} \beta_n = 0$$

$$c_{m1} \beta_1 + c_{m2} \beta_2 + \dots + c_{mn} \beta_n = 0$$

($\therefore$ This become individually zero).

$$\stackrel{\text{if}}{\Rightarrow} c_{11} = c_{12} = \dots = c_{1n} = 0$$

...

$$c_{m1} = c_{m2} = \dots = c_{mn} = 0$$

$\therefore B$ is L.I

$$\text{Let } T \in L(U, V)$$

$$\Rightarrow T(a_1 d_1 + a_2 d_2 + \dots + a_n d_n) =$$

$$a_1 (c_{11} \beta_1 + c_{12} \beta_2 + \dots + c_{1n} \beta_n)$$

$$+ a_2 (c_{21} \beta_1 + c_{22} \beta_2 + \dots + c_{2n} \beta_n)$$

...

$$+ a_m (c_{m1} \beta_1 + c_{m2} \beta_2 + \dots + c_{mn} \beta_n)$$

$$T_{ij}(A) = a_{ij} P_1 \quad T_{ij}(A) = a_{ij} P_1$$

$$T_{12}(A) = a_{12} P_2$$

$$\Rightarrow T(a_1 A_1 + a_2 A_2 + \dots + a_m A_m) =$$

$$\left(\begin{array}{l} c_{11} T_{11} + c_{12} T_{12} + \dots + c_{1m} T_{1m} \\ + c_{21} T_{21} + c_{22} T_{22} + \dots + c_{2m} T_{2m} \\ \vdots \\ + c_{m1} T_{m1} + c_{m2} T_{m2} + \dots + c_{mm} T_{mm} \end{array} \right) \quad (*)$$

$$\Rightarrow B \text{ spans } L(U, V)$$

$$\Rightarrow B \text{ is the Basis of } L(U, V)$$

$$\therefore B_{L(U, V)} = \{ T_{ij} \mid 1 \leq i \leq n, 1 \leq j \leq m \}$$

$$\therefore \dim(L(U, V)) = mn$$

Ques) What will be the dimension of the vector space which set of all L.T from $M_3(\mathbb{R})$ to $P_3(\mathbb{R})$ i.e. $L(M_3(\mathbb{R}), P_3(\mathbb{R}))$

Ans

$$\dim L(M_3(\mathbb{R}), P_3(\mathbb{R}))$$

$$= \dim(M_3(\mathbb{R})) \times \dim(P_3(\mathbb{R}))$$

$$= 3^2 \times 3 + 1$$

$$= 9 \times 4 = 36$$

Ques) If standard ordered basis of $P^3(\mathbb{R})$ and $P(\mathbb{R})$ are taken as to basis of ordered basis of $L(P^3(\mathbb{R}), P(\mathbb{R})) = \text{Hom}(P^3(\mathbb{R}), P(\mathbb{R}))$

Then we obtained Basis = $\{f_1, f_2, f_3\}$

Then find co-ordinate of L.T

$$f(x, y, z) = x - y + 2z$$

$$\text{Standard Basis of } \mathbb{R}^3 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$\text{standard Basis of } \mathbb{R}^3 = \{1, 1, 1\}$$

$$\therefore f_1(1, 0, 0) = 1 \Rightarrow$$

$$f_1(0, 1, 0) = 0$$

$$f_1(0, 0, 1) = 0$$

$$f: \mathbb{R}^3 \rightarrow \mathbb{R} \text{ in } f(x, y, z) = ax + by + cz$$

$$f_1(1, 0, 0) = 1 \Rightarrow a = 1$$

$$f_1(0, 1, 0) = 0 \Rightarrow b = 0$$

$$f_1(0, 0, 1) = 0 \Rightarrow c = 0$$

$$\therefore f_1(x, y, z) = x$$

$$f_2(1, 0, 0) = 0 \Rightarrow a = 0$$

$$f_2(0, 1, 0) = 1 \Rightarrow b = 1$$

$$f_2(0, 0, 1) = 0 \Rightarrow c = 0$$

$$\therefore f_2(x, y, z) = y$$

$$f_3(1, 1, 0) = 0 \Rightarrow a = 0$$

$$f_3(0, 1, 0) = 0 \Rightarrow b = 0$$

$$f_3(0, 0, 1) = 1 \Rightarrow c = 1$$

$$\therefore f_3(x, y, z) = z$$

$$f = a_1 f_1 + a_2 f_2 + a_3 f_3 = (x-y+2z)$$

$$\therefore (a_1, a_2, a_3) = (1, -1, 2)$$

$$\Rightarrow a_1 x + a_2 y + a_3 z = (x-y+2z)$$

$$\therefore c_1 = 1, c_2 = -1, c_3 = 2$$

$$\therefore (c_1, c_2, c_3) = (1, -1, 2) \rightarrow \text{is co-ordinates of L.T}$$

$$f(\text{img } T) = x-y+2z$$

(Q.10) If T is linear operator on vector space V such that $T^2(v) = 0 \ ; \ \forall v \in V$ then which of the following must be true.

① $\text{Range}(T) = \text{Nullity}(T) \mid \text{Null space}(T)$

② $\text{Range}(T) \subseteq \text{Nullity}(T) \mid \text{Null space}(T)$

③ $\text{Range}(T) \supseteq \text{Nullity}(T) \mid \text{Null space}(T)$

④ At least one of the above is correct

Soln:-

$$T^2(v) = 0 \ ; \ \forall v \in V$$

$$\Rightarrow T(T(v)) = 0$$

$$\Rightarrow T(v) \in N(T) \ ; \ \forall v \in V$$

$$\therefore R(T) \subseteq N(T)$$

* MATRIX OF a LINEAR TRANSFORMATION

- ↳ If T is L.T from $V(F)$ to $V(F)$
 i.e. $T: V(F) \rightarrow V(F)$, then w.r.t ordered
 basis of U & V as

$$B_U = \{u_1, u_2, \dots, u_n\} \quad \text{[By default this is not given then we take standard basis]}$$

$$B_V = \{v_1, v_2, \dots, v_m\}$$

Transformation matrix denoted by

$[T] = T = [T; B_U, B_V]$ here it's j^{th}
 column as Transpose of the co-ordinate
 of $T(u_j)$ w.r.t ordered basis of V .

- ↳ If $T(u_j) = a_{1j}v_1 + a_{2j}v_2 + \dots + a_{mj}v_m$
 then co-ordinate is $(a_{1j}, \dots, a_{mj})$

i.e.

$$T = \begin{matrix} & & T(u_1) & \dots & T(u_j) & \dots & T(u_n) \\ \begin{matrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{matrix} & \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} & \dots & \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} & \dots & \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} \end{matrix}$$

co-ordinate of $T(u_j)$ column.

(Quc) Consider L.T $T: P_3(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ given by

$T(p(x)) = p(3x+5)$ find transformation
 matrix w.r.t ordered basis $\{1, x, x^2, x^3\}$
 for both P_3 .

$$p_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

$$T(p(x)) = p(3x+5)$$

put $p(x) = 1$ then $p(3x+5) = 1$

put $p(x) = x$ then $p(3x+5) = 3x+5$

put $p(x) = x^2$ then $p(3x+5) = (3x+5)^2$

$$\therefore T(x^2) = (3x+5)^2 = 9x^2 + 30x + 25$$

put $p(x) = x^3$ then $p(3x+5) = (3x+5)^3$

$$\therefore T(x^3) = (3x+5)^3 = (9x^2 + 30x + 25)(3x+5)$$

$$\therefore \begin{matrix} & T(1) & T(x) & T(x^2) & T(x^3) \\ \begin{matrix} 1 \\ x \\ x^2 \\ x^3 \end{matrix} & \begin{bmatrix} 1 & 5 & 25 & 125 \\ 0 & 3 & 30 & 225 \\ 0 & 0 & 9 & 135 \\ 0 & 0 & 0 & 27 \end{bmatrix} \end{matrix}$$

Ques) If $T: P_3(\mathbb{R}) \rightarrow P_4(\mathbb{R})$ is L.T given by

$$T(p(x)) = p'(x) + \int_0^x p(x) dx \text{ then if } A \text{ is}$$

Transformation matrix w.r.t ordered basis $\{1, x, x^2, x^3\}$ & $\{1, x, x^2, x^3, x^4\}$ respectively of $P_3(\mathbb{R})$ & $P_4(\mathbb{R})$, then

1) $a_{23} = 2$

2) $a_{33} = 0$

solⁿ:

	$T(1)$	$T(x)$	$T(x^2)$
1			0
x			2
x^2			0
x^3			$\frac{2}{3}$

$$T(x^2) = 2x + \frac{x^3}{3} \quad \text{as } T(p(x)) = \int_0^x p(t) dt + \int_0^x p^{(1)}(t) dt$$

(Ques) consider a transformation $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$
 is $T(X) = AX$ where A is given matrix
 then trace of transformation matrix = ?

1) $\text{tr}(A)$ 2) $2\text{tr}(A)$ 3) $4\text{tr}(A)$ 4) none

solⁿ:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

and standard basis of $\mathbb{R}^2 = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	a	0	b	0
$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	0	a	0	b
$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	c	0	d	0
$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	0	c	0	d

$$T \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$T \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix} = a \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$T \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix} = b \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$T \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix} = b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\therefore \text{tr}(A) = a + d \quad \text{trace} = a + d + d = d(a + d)$$

$$\therefore \text{tr}(T) = a \text{tr}(A)$$

(Que) If $T: M_n(A) \rightarrow M_n(A)$ is L.T. given by $T(X) = AX$, where $A = [a_{ij}]_{n \times n}$ is given matrix and T is Transformation matrix. Then $\text{tr}(T) =$

- (1) $\text{tr}(A)$ (2) $n \text{tr}(A)$ (3) $(n-1) \text{tr}(A)$
(4) $n^2 \text{tr}(A)$

soln

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} 0 & \dots & 0 \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ \vdots \\ a_{nn} \end{bmatrix}$$

$$\text{If } \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} 0 & \dots & 1 \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ a_{nn} \end{bmatrix}$$

$$\text{Ag } \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} 0 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} = \begin{bmatrix} a_{12} & \dots \\ \vdots & \vdots \\ \vdots & \vdots \end{bmatrix}$$

$$\text{If do so on } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$\xrightarrow{\text{nth times}} a_{nn}$
 $\xrightarrow{\text{nth times}} a_{nn}$
 $\xrightarrow{\text{nth times}} a_{nn}$

$$= \text{tr}(A) = n \cdot \text{tr}(M)$$

(Case) If $T: \mathbb{R}^2(A) \rightarrow \mathbb{R}^2(A)$ and Transformation matrix $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ w.r.t to ordered basis

$\{e_1, e_2\}$ & $\{e_1, e_2\}$ respectively

then $T(e_1) = \underline{\hspace{2cm}}$

$$\text{Sol: } \{e_1, e_2\} = \{a_1, a_2\} \text{ & } A = \begin{bmatrix} p_1 & p_2 \\ p_1 & p_2 \end{bmatrix}$$

$$\{e_1, e_2\} = \{b_1, b_2\}$$

4th step:-

$$\text{Let } (x, y) = a \overset{d_1}{(1, 1)} + b \overset{d_2}{(1, 0)}$$

$$\therefore a + b = x$$

$$a = y$$

$$\Rightarrow b = x - y$$

$$\begin{aligned} \therefore T(x, y) &= y \overset{d_1}{T(1, 1)} + b \overset{d_2}{T(1, 0)} \\ &= y T(1, 1) + (x - y) T(1, 0) \\ &= y [1 \cdot \beta_1 + 1 \cdot \beta_2] + x - y [2\beta_1 + 3\beta_2] \\ &= y [(1\beta_1) + 1\beta_2] + x - y [2(1\beta_1) + 3(1\beta_2)] \\ &= y [-1, 1] + x - y (8, 7) \\ &= (8x - 9y, 7x - 6y) \end{aligned}$$

Ques If $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is L.T such that $T(a) = a \cdot a \cdot \forall a \in \mathbb{R}^n$ & $a \neq 0$. then Transformation matrix of T is

Solⁿ Let ordered basis $d_1, d_2, \dots, d_n$

Transformation matrix is

$$\begin{matrix} & T(d_1) & T(d_2) & \dots & T(d_n) \\ \begin{matrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{matrix} & \begin{bmatrix} a & 0 & & 0 \\ 0 & a & & 0 \\ 0 & 0 & \dots & 0 \\ & 0 & & a \end{bmatrix} \end{matrix}$$

(Ques) If $d_1, d_2, d_3, \dots, d_n$ are L.I. Eigen vectors of transformation matrix of L.T

$S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ then transformation matrix must be diagonal matrix, true or false, w.r.to $d_1, d_2, d_3, \dots, d_n$

Ans:- $d_1, d_2, \dots, d_n$ are L.I. eigen vectors.

$$\therefore A d_i = \lambda_i d_i$$

$$\Rightarrow T(d_i) = \lambda_i d_i$$

then transformation matrix is

$$\begin{matrix} T(d_1) & T(d_2) & \dots & T(d_n) \\ \begin{matrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{matrix} \end{matrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Any L.C of L.I. as diagonal matrix is again diagonal matrix i.e. eg C, I, T , S & T all are diagonal matrix.

(*) $\rightarrow$ If T is linear operator on $V(F)$ then linear operator is Nilpotent / Idempotent / Identity / diagonal according as its transformation matrix satisfy those property.

$$T.M = \begin{matrix} & T(d_1) & T(d_2) \\ \begin{matrix} d_1 \\ d_2 \end{matrix} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{matrix}$$

$$T(d_1) = a d_1 + c d_2$$

$$T(d_2) = b d_1 + d d_2$$

$$TOT(d_1) = a T(d_1) + c T(d_2)$$

$$= a [a d_1 + c d_2] + c [b d_1 + d d_2]$$

$$= (a^2 + cb) d_1 + (ac + cd) d_2$$

$$TOT(d_2) = b T(d_1) + d T(d_2)$$

$$= b [a d_1 + c d_2] + d [b d_1 + d d_2]$$

$$= (ab + db) d_1 + (bc + d^2) d_2$$

$$\text{(terms formation matrix)} = \begin{matrix} & \begin{matrix} T^2(d_1) & T^2(d_2) \end{matrix} \\ \begin{matrix} d_1 \\ d_2 \end{matrix} & \begin{bmatrix} a^2 + cb & ab + db \\ ac + cd & d^2 + bc \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} & \begin{matrix} TOT(d_1) & TOT(d_2) \end{matrix} \\ \begin{matrix} d_1 \\ d_2 \end{matrix} & \begin{bmatrix} a^2 + cb & ab + db \\ ac + cd & d^2 + bc \end{bmatrix} \end{matrix}$$

- (A) trace of T is zero (B) trace of T^2 is zero
 (C) multiply up T is T (D) $T^{n+1} = 0$ if n is odd.
 (E) $T^2 = 0$ if n is even.

→ If T is transformation matrix of linear operators of S on V then T^k will be transformation matrix of linear operators of S on V .

(Case) If T is linear operator on vector space V of n basis $p = \{e_1, e_2, \dots, e_n\}$ such that

$T(e_i) = e_{i+1}$ and $e_{n+1} = 0$. Then T^k is zero map where k is

- (1) n (2) $n-1$ (3) n (4) n

The transformation matrix is

	$T(e_1)$	$T(e_2)$	$\dots$	$T(e_n)$	
e_1	0	0	0	0	Here transformation matrix is strictly lower triangular. then $A^n = 0$ $A \neq 0$ $A^n = 0$.
e_2	1	0	0	0	
e_3	0	1	0	0	
$\vdots$					
e_n	0	0	1	0	

(Case $n=0$)

Note: If (x) & $m(x)$ of Transformation matrix is given then on the basis of this information transformation matrix is written as Jordan canonical form.

(Ques) If T is linear operator on vector space V such that $C(x) = x^3(x-1)^2$ & $m(x) = x^2(x-1)$ of the transformation matrix then which of the followings are true.

- ✓ 1) T has 2 Jordan Blocks
- ✓ 2) T has 4 Jordan Blocks
- ✓ 3) operator T induced on $\frac{V}{\ker(T)}$ is Nilpotent.
- ✗ 4) operator T induced on $\frac{V}{\ker(T-I)}$ is Nilpotent
- ✓ 5) operator T induced on $\frac{V}{\ker(T-I)}$ is Nilpotent.

Sol.:- Transformation matrix is

$$T = \begin{bmatrix} 0 & 1 & & & \\ & 0 & & & \\ & & 0 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

As $C(x) = x^3(x-1)^2$
 & $m(x) = x^2(x-1)$

∴ T has 4 Jordan blocks.

∴ $\{d_1, d_2, d_3, d_4, t\}$ is basis of T

$$T = \begin{matrix} & \begin{matrix} T(d_1) & T(d_2) & T(d_3) & T(d_4) & T(d_5) \end{matrix} \\ \begin{matrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \end{matrix} & \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & & & \\ & & 0 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix} \end{matrix}$$

$$\therefore T(d_1) = 0, \quad T(d_2) = d_1$$

$$T(d_3) = 0, \quad T(d_4) = d_4, \quad T(d_5) = d_5$$

$$\therefore \ker(T) = L[d_1, d_3] = W$$

$$\therefore \frac{B_W}{W} = \{w+d_2, w+d_4, w+d_5\}$$

③ Now apply T on $\frac{W}{W}$ Basis of $\frac{W}{W}$

$$\therefore T: \frac{W}{W} \rightarrow \frac{W}{W}$$

$$T = \begin{matrix} & \begin{matrix} T(w+d_2) & T(w+d_4) & T(w+d_5) \end{matrix} \\ \begin{matrix} w+d_2 \\ w+d_4 \\ w+d_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$\therefore T(w+d_2) = T(w) + T(d_2)$$

$$= w + d_1 = w = 0 \quad ; \quad (-d_1 \in W \text{ as } d_1 = 0 \text{ in } W = 0)$$

$\therefore$ 'T' has a e.g. in T $\Rightarrow$ T is not nilpotent
 bcoz nilpotent matrix has all ev in 0.

$$\begin{aligned} 4) \quad (T - 5I)(a_1) &= T(a_1) - 5a_1 \\ &= -5a_1 \quad (\because T(a_1) = 0) \end{aligned}$$

$$\begin{aligned} (T - 5I)(a_2) &= T(a_2) - 5a_2 \\ &= a_1 - 5a_2 \quad (\because T(a_2) = a_1) \end{aligned}$$

$$\begin{aligned} (T - 5I)(a_3) &= T(a_3) - 5a_3 \\ &= -5a_3 \quad (\because T(a_3) = 0) \end{aligned}$$

$$\begin{aligned} (T - 5I)(a_4) &= T(a_4) - 5a_4 \\ &= a_4 - 5a_4 = -4a_4 \quad (\because T(a_4) = a_4) \end{aligned}$$

$$\begin{aligned} (T - 5I)(a_5) &= T(a_5) - 5a_5 \\ &= a_5 - 5a_5 \\ &= -4a_5 \quad (\because T(a_5) = a_5) \end{aligned}$$

$$5) \quad (T - I)(a_1) = -a_1$$

$$(T - I)(a_2) = a_1 - a_2$$

$$(T - I)(a_3) = -a_3$$

$$(T - I)(a_4) = a_4 - a_4 = 0$$

$$(T - I)(a_5) = a_5 - a_5 = 0$$

$$\therefore \text{Ker}(T) = \text{span}\{a_4, a_5\}$$

$$\ker(T-I) = \{d_1, d_2\} = W$$

$\therefore B_W = \{w+d_1, w+d_2, w+d_3\}$ then T is

$$T = \begin{matrix} & \begin{matrix} T(w+d_1) & T(w+d_2) & T(w+d_3) \end{matrix} \\ \begin{matrix} w+d_1 \\ w+d_2 \\ w+d_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$T(w+d_1) = T(w) + T(d_1)$$

$$= w + 0 = 0 \quad (\because w=0 \text{ in invariant space})$$

$$T(w+d_2) = T(w) + T(d_2)$$

$$= w + d_1 \quad (\because T(d_2) = d_1)$$

$$T(w+d_3) = T(w) + T(d_3)$$

$$= w + 0 \quad (\because T(d_3) = 0)$$

$\therefore T$ is Nilpotent ^{with} index 2

$$\text{as } T(T(w+d_2)) = T(w+d_1) \quad (\because T^2 = TT) \\ = w + 0 \\ = 0$$

$$\therefore T \circ T \left(\frac{N}{\ker(T-I)} \right) = \text{zero map.}$$

$$z = x + iy \\ = (x, y) \\ = \mathbb{R}^2(\mathbb{R})$$

$$T: z \rightarrow ze^{i\theta} : \text{anti-clockwise rotate}$$

$$T: z \rightarrow ze^{-i\theta} : \text{clockwise rotate}$$

$$T: z \rightarrow ze^{i\theta} \rightarrow \text{rotate in anti-clockwise direction}$$

$$T(x+iy) = (x+iy)(\cos\theta + i\sin\theta) \\ = x\cos\theta - y\sin\theta + i(y\cos\theta + x\sin\theta)$$

Note:

$T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ is L.T which rotates every vector of $\mathbb{R}^2(\mathbb{R})$ by an angle θ in an anti-clockwise direction.

$$T(x, y) = (x\cos\theta - y\sin\theta, y\cos\theta + x\sin\theta)$$

(Quc) Find the transformation matrix w.r.t. ordered basis $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{R}^2(\mathbb{R})$ for the transformation which rotates every vector of $\mathbb{R}^2$ by an anticlockwise.

Sol.

Transformation matrix is

$$\begin{bmatrix} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} & T \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

$$T(x, y) = T \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\therefore T(x, y) = (x\cos\theta - y\sin\theta, y\cos\theta + x\sin\theta)$$

$$= (x \cos \theta + y \sin \theta, y \cos \theta - x \sin \theta)$$

$$\therefore T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos \theta + y \sin \theta \\ -x \sin \theta + y \cos \theta \end{bmatrix}$$

$$\therefore T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}$$

$$= \sin \theta \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \cos \theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \cos \theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \sin \theta \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

(Ans) If T is L.T which transform every vector of $\mathbb{R}^2$ in such a way that its x -coordinates is scaled by a factor of $\frac{1}{2}$ & y co-ordinates is scaled by factor of 2 then its Transformation matrix w.r.t basis $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

$$T(x, y) = \left(\frac{1}{2}x, 2y \right)$$

$$\text{Let } \begin{bmatrix} x \\ y \end{bmatrix} = n \begin{bmatrix} 1 \\ 1 \end{bmatrix} + m \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow l+m=a \quad \& \quad a-m=b$$

$$\Rightarrow \quad l = \frac{a+b}{2} \quad \& \quad m = \frac{a-b}{2}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 2 \end{bmatrix} \quad \& \quad T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -2 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad T \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

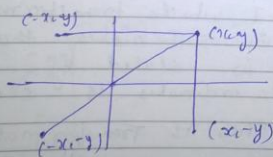
$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} ; \quad l = \frac{a+b}{2} = \frac{1/2 + 2}{2} = 5/4$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} \quad m = \frac{a-b}{2} = \frac{1/2 - 2}{2} = -3/4$$

(Quc) write down the transformation which reflects every vector of $\mathbb{R}^2$ ~~through~~ ^{about}

- (1) x -axis (2) y -axis (3) origin (4) line $y=x$

points



$$(1) \quad T(x, y) = (x, -y) \quad \text{about } x\text{-axis}$$

$$(2) \quad T(x, y) = (-x, y) \quad \text{about } y\text{-axis}$$

$$(3) \quad T(x, y) = (-x, -y) \quad \text{about origin}$$

4) $T(x, y) = (y, x)$: line $y=x$

Ques) find a transformation on $\mathbb{R}^3(\mathbb{R})$ which reflects each of its vector about origin and then find transformation which project every vector orthogonally on xz plane.

Solⁿ:- $\mathbb{R}^3(\mathbb{R})$ which reflects each of its vector about origin is

$$T(x, y, z) = (-x, -y, -z)$$

and transformation which project every vector orthogonally on xz plane is

$$T(x, y, z) = (x, 0, z)$$

Ques) If T is linear operator i.e. $T: \mathbb{R}^n(\mathbb{R}) \rightarrow \mathbb{R}^n(\mathbb{R})$ then which of the following may be possible

1) Range & kernel of T are straight line through origin

2) Range is plane & ker is straight line

3) Range is straight line & ker is plane

4) None

<u>Rank</u>	Rank (T)	Nullity (T)	Range (T)	ker (T)
2	2	0	2D space	origin (0,0,0)
1	1	1	line through origin	line through origin
0	0	2	(0,0) origin point	2D-space

$S = \{ (x, y) \mid x + 2y = 0 \} : a, b \neq 0$
 $\downarrow$
 eqn of a line in 2D plane through origin
 of dim 1

Ques) T is linear operator i.e. $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$
 then by Rank Nullity theorem we get the following table.

Rank (T)	Nullity (T)	Range (T)	ker (T)
3	0	3D space	point (0,0,0)
2	1	plane passing through origin	line passing through origin
1	2	st. line passing through origin	plane passing through origin
0	3	point (0,0,0)	3D-space

$d(x, y, z) \mid x + 2y + z = 0$ is the plane passing through origin

$$\{(x, y, z) \mid \begin{matrix} ax+by+cz=0 \\ a_1x+b_1y+c_1z=0 \end{matrix} \} \text{ which is}$$

a straight line passing through origin

$$\frac{ax}{bc_1 - cb_1} = \frac{y}{ac_1 - ca_1} = \frac{z}{ab_1 - ba_1}$$

(Que) If T is linear operator on $\mathbb{R}^3$ given by $T(x) = Ax$ where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ then

1) Range(T) is plane and Ker(T) is st. line

2) " " st. line & " " plane

3) " " st. line & " " st. line

4) " " plane & " " plane

$$\text{Sol: } T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Rank}(A) = 2 \text{ then Rank}(T) = 2$$

$$\therefore \text{Nullity}(T) = 1$$

$\therefore \text{Rank}(T) = 2$ the Range(T) is plane

$\therefore \text{Nullity} = 1$ then Ker(T) is st. line

* Linear functional:

If $V(F)$ is a vector space then as $F(F)$ is vector space so $\exists$ linear transformation from $V(F)$ to $F(F)$. Such linear transformation is known as "Linear functional".

Defn: If $V(F)$ is vector space then transformation from $V(F) \rightarrow F(F)$ are known as "linear functional" or "scalar transformation" on $V(F)$.

* Example of Linear Transformation

1) $T: P^3(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$ then

$$T(x, y, z) = ax + by + cz$$

(Set of all real function from $P^3(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$)

2) $T: M_n(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$ then

$$T(A) = \sum_{j=1}^n \sum_{i=1}^n a_{ij} a_{ji}$$

(Set of all real matrices from $M_n(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$)

$$\text{also } T(A) = \text{tr}(A)$$

3) $T: P_3(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$ then

$$T(a + bx + cx^2 + dx^3) = d_1 a + d_2 b + d_3 c + d_4 d$$

$\therefore d_1, d_2, d_3, d_4 \in \mathbb{R}$

* Set of all linear functional on vector space $V(F)$ form a vector space in case of dual space. w.r.t
 $(f_1 + f_2)(a) = f_1(a) + f_2(a)$ & $(cf)(a) = c f(a)$. It is denoted by V^* .

* Dual space and dual basis of a vector space $V(F)$

If $V(F)$ is a vector space then set of all linear functional on $V(F)$ form a vector space w.r.t

$$(f_1 + f_2)(a) = f_1(a) + f_2(a)$$

$$(cf)(a) = c f(a)$$

which is also known as Dual space or conjugate space of vector space V .

It is denoted by,

$$V' = \hat{V} = V^* = \{ f \mid f \text{ is linear functional on } V \}$$

→ Any basis of dual space is known as Dual basis.

Also, if $\dim V = n$ then $\dim V' = \dim V$
 $= n$
 $= n$

(Ques) find the dual basis of $P_2(\mathbb{R})$ w.r.t its ordered basis $\{1, 1+x, 1+x+x^2\}$ if $\{f_1, f_2, f_3\}$ is the dual basis then find $\{a_1, a_2, a_3\}$ such that

$$f(1+x^2+x^3) = 2a_1 + 3a_2 + 4a_3 \quad \text{and}$$

$$f = a_1 f_1 + a_2 f_2 + a_3 f_3$$

$$\{1, 1+x, 1+x+x^2\} = 1(1, 0, 0) + (1+x)(0, 1, 0) + (1+x+x^2)(0, 0, 1)$$

Ans: Let $f_1(a+bx+cx^2) = l_1 a + l_2 b + l_3 c$

$$f_1(1) = l_1 = 1$$

$$f_1(1+x) = l_1 + l_2 = 0$$

$$f_1(1+x+x^2) = l_1 + l_2 + l_3 = 0$$

$$\therefore l_1 = 1, \quad l_2 + l_3 = 0 \quad \& \quad l_1 + l_2 + l_3 = 0$$

$$\Rightarrow l_2 = -1, \quad \& \quad l_3 = 0.$$

$$\boxed{f_1(a+bx+cx^2) = a-b}$$

Now

$$f_2(1) = l_1 = 0$$

$$f_2(1+x) = l_1 + l_2 = 1$$

$$f_2(1+x+x^2) = l_1 + l_2 + l_3 = 0$$

$$\Rightarrow l_1 = 0, \quad l_2 = 1, \quad l_3 = -1$$

$$\therefore \boxed{f_2(a+bx+cx^2) = b-c}$$

Now

$$f_3(1) = l_1 = 0$$

$$f_3(1+x) = l_1 + l_2 = 0$$

$$f_3(1+x+x^2) = l_1 + l_2 + l_3 = 1$$

$$\therefore l_1 = 0, \quad l_2 = 0, \quad l_3 = 1$$

$$\boxed{f_3(a+bx+cx^2) = c}$$

* Set of all linear functional from V which transformed on zero space is called "annihilator" of subset of S of V . It is denoted by S^0 and $S^0 = \{f \in V' / f(x) = 0; \forall x \in S\}$

Let $f = d_1 f_1 + d_2 f_2 + d_3 f_3$

$$f(a+bx+cx^2) = (d_1 f_1 + d_2 f_2 + d_3 f_3)(a+bx+cx^2)$$

$$\therefore 9a+3b+4c = d_1(a-b) + d_2(b-c) + d_3(c) \quad ; \forall a, b, c$$

$$\Rightarrow d_1 = 9, \quad d_2 - d_1 = 3 \quad \& \quad d_3 - d_2 = 4$$

$$\Rightarrow d_2 = 5, \quad d_3 = 9$$

$$\therefore (d_1, d_2, d_3) = (9, 5, 9)$$

* Annihilator of a subset S of V .

If $V(V)$ is vector space and S is subset of V ($\because S \subseteq V$) then Annihilator of S denoted by S^0

$$S^0 = \{f \in V' / f(x) = 0; \forall x \in S\}$$

Th

$\Rightarrow$ If S is subset of V then S^0 is subspace of V'

Proof:

$$f_1, f_2 \in S^0 \Rightarrow f_1(x) = 0; \forall x \in S$$

$$f_2(x) = 0; \forall x \in S$$

Let $a, b \in \mathbb{R}$ then

$$(af_1 + bf_2)(x) = af_1(x) + bf_2(x)$$

$$= a \cdot 0 + b \cdot 0$$

$$= 0 \quad \forall x \in S;$$

$$\therefore af_1 + bf_2 \in S^0$$

then S^0 is subspace of V'

Theorem:

If W is subspace of vector space $V(F)$ then

$$\dim W + \dim W^\circ = \dim V$$

Proof:

$$\text{Let } B_W = \{d_1, d_2, \dots, d_m\}$$

Now $B_W = \{d_1, d_2, \dots, d_m, p_1, p_2, \dots, p_n\}$
converted into dual basis

$$B_{W^\circ} = \{f_1, f_2, f_3, \dots, f_m, f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$$\therefore B = \{f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$\therefore B$ is L.I

Now, let $f \in W^\circ$

$$\Rightarrow f(c_1 d_1 + c_2 d_2 + \dots + c_m d_m) = 0 \quad \forall c_i$$

$$\Rightarrow (a_1 f_1 + a_2 f_2 + \dots + a_m f_m + b_1 f_{m+1} + \dots + b_n f_{m+n})$$

$$(c_1 a_1 + c_2 a_2 + \dots + c_m a_m) = 0$$

$$\Rightarrow a_1 c_1 + a_2 c_2 + \dots + a_m c_m = 0$$

$$\Rightarrow a_1 = a_2 = \dots = a_m = 0 \quad ; \quad \forall c_i$$

$$\therefore f = b_1 f_{m+1} + b_2 f_{m+2} + \dots + b_n f_{m+n}$$

$\therefore B$ spans W°

$$\Rightarrow L(B) = W^\circ$$

* set of all functional which is annihilator of V and which is isomorphic to zero space is called annihilator of annihilator of subset S of vector space V is denoted by $S^{\circ\circ}$ and defined by $S^{\circ\circ} = \{ f \in V^* \mid f(v) = 0, \forall v \in V \}$

$$\therefore B_{W^0} = \{ f_{m+1}, f_{m+2}, \dots, f_{m+n} \}$$

$$\therefore \dim W + \dim W^0 = \dim V$$

$$\therefore m + n = m + n$$

$$\Rightarrow \dim V = m+n$$

(Ques) If V is v. space of all polynomials of degree upto 100 (i.e. ≤ 100) and W be its subspace given by

$$W = \{ p(x) : D^{50} p(x) = 0 \}$$

$$\text{then } \dim W^0 = ?$$

Solⁿ:-

$$\dim W + \dim W^0 = \dim V$$

$$\therefore \dim W^0 = \dim V - \dim W$$

$$= 101 - 50$$

$$= 51$$

$$W = \{ p(x) : D^{50} p(x) = 0 \}$$

$$= \{ a_0 + a_1 x + a_2 x^2 + \dots + a_{49} x^{49} \}$$

$$\therefore \dim W = 50$$

* Annihilator of Annihilator of a subset S of vector space V .

$\rightarrow$ It is denoted by $S^{\circ\circ}$

$$\therefore S^{\circ\circ} = \{ f \in V^* \mid f(v) = 0, \forall f \in S^0 \}$$

$$= \{ f \in V^* \mid f(v) = 0, \forall v \in V \}$$

Result

If W is subspace of vector space $V(F)$
then $W^{00} = W$

Proof:

$$\text{Let } B_W = \{w_1, w_2, \dots, w_m\}$$

$$B_V = \{d_1, d_2, \dots, d_m, \beta_1, \beta_2, \dots, \beta_n\}$$

$$B_{W'} = \{f_1, f_2, \dots, d_m, f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$$B_{W^{00}} = \{f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$$B_{W^{00}} = \{f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$$B_{W^{00}} = \{d_1, d_2, \dots, d_m\}$$

$$\therefore W = W^{00} = W^{0000} = W^{000000} \dots$$

all are same basis

put $w = w_0$ then

$$w^0 = w^{00} = w^{0000} = w^{000000} = \dots$$

all are same basis

(Ques) If S is Subspace of $V(F)$ then

$$S \subseteq S^{00}$$

$$(1) = (2) \subseteq (3) \supseteq$$

Proof:

$$S = \{d_1, d_2, \dots, d_m\}$$

$$S^0 = \{f_{m+1}, f_{m+2}, \dots, f_{m+n}\}$$

$$S^{00} = \{d_1, d_2, \dots, d_m\} = L(S)$$

$$\therefore L(S) = S^{00}$$

29/03/2015

morning

* Invariance:-

If $V(F)$ is a vector space & T is linear operator on V then a subspace W of V is said to be "invariant" under T if

$$v \in W \Rightarrow T(v) \in W$$

$$\text{i.e. } T(W) \subseteq W$$

(Ques) consider a vector space $\mathbb{R}^3$ and linear operator $T(x, y, z) = (x+y, 2x+y, z)$ then which of the following are subspaces are invariant?

(1) $\mathbb{R}^3$

(2) Zero space

(3) $\{(a, b, 0) \mid a, b \in \mathbb{R}\}$

(4) $\{(a, 0, 0) \mid a \in \mathbb{R}\}$

(5) $\{(0, 0, c) \mid c \in \mathbb{R}\}$

Note $V(F)$ & Zero space are always invariant under T .

(3) $T(a, b, 0) = (a+b, 2a+b, 0) \in \{(a, b, 0) \mid a, b \in \mathbb{R}\}$
 $\therefore$ It is invariant

$$(4) T(a, 0, 0) = (a, 2a, 0) \notin \{(a, 0, 0) | a \in \mathbb{R}\}$$

$\therefore$ then not a invariant.

$$(5) T(0, 0, c) = (0, 0, c) \in \{(0, 0, c) | c \in \mathbb{R}\}$$

$\therefore$ is invariant

* Reducibility:

If $V(E)$ is a vector space and T is linear operator on $V(E)$, then V is said to be reducible by its subspaces W_1 & W_2 under T if,

$$1) V = W_1 \oplus W_2$$

2) W_1 & W_2 are invariant under T .

(Que) Let $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ be a linear operator

$$\text{given by } T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a-b & 2a+3b \\ 2c+3d & c-d \end{bmatrix}$$

w.r.t which pair of subspaces $M_2(\mathbb{R})$ is Reducible under T .

(1) $M_2(\mathbb{R})$ & zero space

$$(2) \left\{ \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \mid a, b \in \mathbb{R} \right\} \text{ & } \left\{ \begin{bmatrix} 0 & 0 \\ c & d \end{bmatrix} \mid c, d \in \mathbb{R} \right\}$$

$$(3) \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \mid a \in \mathbb{R} \right\} \text{ & } \left\{ \begin{bmatrix} 0 & b \\ c & d \end{bmatrix} \mid b, c, d \in \mathbb{R} \right\}$$

(4) None.

sol: (1) Trivial as $M_2(K)$ & Zero space $= V$
and $M_2(K)$ & zero space is invariant under T .

(2) intersection of both $= 0$

Let apply transformation

$$\begin{bmatrix} a-b & 2a+3b \\ a & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ a-b & a-b \end{bmatrix}$$

both are invariant.

(3)

$$T \begin{bmatrix} a & 0 \\ a & 0 \end{bmatrix} = \begin{bmatrix} a & 2a \\ 0 & 0 \end{bmatrix} \text{ is not invariant}$$

* Projection:

If $V(F)$ is a vector space
& W_1 & W_2 are subspaces of $V(F)$ s.t.
 $V = W_1 \oplus W_2$, then every vector $\alpha \in V$ can be
uniquely expressed as

$$\alpha = \alpha_1 + \alpha_2; \text{ where } \alpha_1 \in W_1, \& \alpha_2 \in W_2$$

Then linear operator and L.T $P: V \rightarrow V$

given by $P(\alpha) = \alpha_1$ is projection of V .

upon a vector
along. 0112101

or
2110101

01

upon w_1 and along w_2

Ans) consider $P_3(\mathbb{R})$ and its subspace

$$W_1 = \{1, 1+x, 1+x+x^2\} \text{ and } W_2 = \{1+x+x^2+x^3\}$$

is $v \in P_3(\mathbb{R}) = W_1 \cup W_2$?

solⁿ $a(1) + b(1+x) + c(1+x+x^2) = d(1+x+x^2+x^3)$

$$\therefore a(1) + b(1+x) + c(1+x+x^2) - d(1+x+x^2+x^3) = 0$$

$$\Rightarrow -d=0 \Rightarrow d=0, \quad c=0, \quad b=0, \quad a=0$$

$\therefore$ intersection of w_1 & $w_2 = 0$

$$\therefore \dim W_1 = 3 \text{ \& } \dim W_2 = 1$$

$$\therefore \dim W_1 + \dim W_2 = 4 = \dim P_3(\mathbb{R})$$

$$\therefore P_3(\mathbb{R}) = W_1 \cup W_2$$

$\therefore$ yes

Find linear operator on $P_3(\mathbb{R})$ which is projection of each of the vectors of $P_3(\mathbb{R})$ upon w_1 and along w_2 .

solⁿ Let $a+bx+cx^2+dx^3 = k_1(1) + k_2(1+x) + k_3(1+x+x^2) + k_4(1+x+x^2+x^3)$

$\therefore$ upon w_1 i.e. 2110101 & along w_2 i.e. 0112101

$k_1, k_2, \& k_3$ find easily

$$\therefore e_4 = d$$

$$l_3 + l_4 = c \Rightarrow l_3 = c - d$$

$$l_2 + l_3 + l_4 = b \Rightarrow l_2 = b - c$$

$$l_1 + l_2 + l_3 + l_4 = a \Rightarrow l_1 = a - b$$

$\therefore$ linear transformation $p: P_3(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ is

$$p(a + bx + cx^2 + dx^3) = (a-b) + (b-c)x + (c-d)(1+x+x^2)$$

$$\therefore p(a + bx + cx^2 + dx^3) = (a-b) + (b-c)x + (c-d)(1+x+x^2)$$

(ex) If $v = w_1 \oplus w_2$ & $p: V \rightarrow V$ is a projection upon w_1 along w_2 then $p(w_1) = w_1$.

Proof:

$$p \cdot v \rightarrow v$$

$$p(v) = p(w_1 + w_2) = p(w_1 + 0) = w_1$$

$$\text{then } p(w_1) = w_1$$

($\therefore$ upon w_1 & along w_2)
 $\therefore w_2 \rightarrow 0$ in w_2)

$$\therefore p^2(w_1) = w_1$$

$$\text{also } p(v) = w_1 \Rightarrow p(p(w_1)) = p^2(v) = w_1$$

$$p(p(v)) = p(w_1)$$

$$\therefore p^2(v) = w_1$$

Th¹ A linear operator p on vector space V is projection iff p is idempotent.

Proof part is above.

(Que) p_1 and p_2 are projection on vector space $V(F)$, then $p_1 + p_2$ is again a projection iff

1) $p_1 = p_2$

2) $p_1 p_2 = 0$

3) $p_1 p_2 = 0$ & $p_2 p_1 = 0$

4) None.

Solⁿ: Let p_1 is projection $\Rightarrow p_1^2 = p_1$ &

p_2 is projection $\Rightarrow p_2^2 = p_2$

$$\begin{aligned}\therefore (p_1 + p_2)^2 &= (p_1 + p_2)(p_1 + p_2) \\ &= p_1^2 + p_1 p_2 + p_2 p_1 + p_2^2 \\ &= p_1 + p_2 + p_1 p_2 + p_2 p_1\end{aligned}$$

$\therefore p_1 + p_2$ is projection iff $p_1 p_2 + p_2 p_1 = 0$.

$$\therefore (p_1 + p_2)^2 = p_1 + p_2 \text{ iff } p_1 p_2 + p_2 p_1 = 0.$$

Taking composition on both side $p_1 p_2$ in post we get.

$$= P_1 P_2 + P_2 P_1 = 0$$

$$= P_1 P_1 P_2 + P_1 P_2 P_1 = 0$$

$$= P_1^2 P_2 + P_1 P_2 P_1 = 0$$

$$= P_1 P_2^2 + P_1 P_2 P_1 = 0 \rightarrow \textcircled{1}$$

or

$$P_2 P_1 P_2 + P_2 P_2 P_1 = 0$$

$$P_2 P_1 P_2 + P_2^2 P_1 = 0$$

$$P_2 P_1 P_2 + P_2 P_1 = 0 \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$ we have

$$P_1 P_2 = 0 \text{ \& \> } P_2 P_1 = 0.$$

(Quc) If p is a projection of V upon w_1 along w_2 then $I-p$ is a projection? and of V upon w_2 along w_1

proof

$$(I-p)^2 = I^2 - 2p + p^2$$

$$= I - 2p + p$$

$$= I - p$$

$\therefore I-p$ is projection

$$\text{also } \alpha = \alpha_1 + \alpha_2$$

$$p(\alpha) = \alpha_1$$

$$\therefore I(\alpha) = \alpha_1 + \alpha_2$$

$$\therefore (I-p)(\alpha) = \alpha_2 \quad \because I-p \text{ is projecting upon } w_2 \text{ along } w_1$$

Ques) Let $T_1: V_1 \rightarrow V_2$ & $T_2: V_2 \rightarrow V_2$ are two linear operators then consider $L = T$
 $T: V_1 \otimes V_2 \rightarrow V_1 \otimes V_2$ given by

$$T(\alpha \otimes \beta) = (T_1(\alpha), T_2(\beta)) ; \forall \alpha \in V_1, \forall \beta \in V_2.$$

Ans) If minimal poly $m_1(x)$ of T_1 is $x^3 + x^2 + x + 1$ and minimal poly $m_2(x)$ of T_2 is $x^4 - x^2 - 2$ then, which of the followings are true.

- 1) Nullity $(T) = 0$.
- 2) Nullity $(T) \neq 0$
- ✓ 3) $\deg$ of m.p of $T = 5$
- 4) $\deg$ (m.p of T) = 7

Proof:

If $m_1(x) = x^3 + x^2 + x + 1$ of T_1

& $m_2(x) = x^4 - x^2 - 2$ of T_2 then

eigenvalue is root of m.p & constant term is non zero then

non singular: T_1 & T_2 is non singular

then T is non singular

$$\text{Nullity}(T) = 0$$

$$\& \text{Nullity}(T_2) = 0$$

$$\text{then Nullity}(T) = 0.$$

$$\therefore T(\alpha, \beta) = (0, 0)$$

$$\Rightarrow T_1(\alpha) = 0 \text{ iff } \alpha = 0$$

$$\Rightarrow T_2(\beta) = 0 \text{ iff } \beta = 0.$$

$$\Rightarrow T(\alpha, \beta) = (T_1(\alpha), T_2(\beta)) \\ = (0, 0)$$

$$\Rightarrow \ker T = (0, 0)$$

$$\therefore \text{Nullity}(T) = 0.$$

Any annihilating poly of T will be simultaneously annihilating poly of T_1 & T_2 .
Annihilating poly of T_2 .

$\therefore$ Annihilating poly of T_1 = multiple of
minimal poly of T_1
 $m_1(x)$

$\therefore$ Annihilating poly of T_2 = multiple of
minimal poly of T_2
 $m_2(x)$

$\Rightarrow$ common multiple of $m_1(x)$ & $m_2(x)$.

$\Rightarrow$ minimal poly of $T = \text{L.C.M of } m_1(x) \text{ \& } m_2(x)$.

$$\Rightarrow x^3 + x^2 + x + 1 = (x^2 + 1)(x + 1)$$

$$x^4 - x^2 - 2 = (x^2 + 1)(x^2 - 1)$$

$$\begin{aligned}\therefore \text{min poly of } T &= \text{L.C.M of } m_1(x) \text{ \& } m_2(x) \\ &= \text{L.C.M} (x+1), (x^2+1), (x^2-2) \text{ } (x^2+1)^2 \\ &= (x+1)(x^2+1)(x^2-2)\end{aligned}$$

$$\therefore \deg(m(x) \text{ of } T) = 5$$

* Inner product Space :-

In this topic we are going to study about vector spaces where field is either real or complex.

Defn: Inner product :-

If $V(F)$ is a vector space then a function from $V \times V$ to F which assigns each ordered pair (α, β) a unique scalar from F such that

- 1) $(\alpha, \beta) = \overline{(\beta, \alpha)}$ (conjugate symmetry)
- 2) $(a\alpha + b\beta, \gamma) = a(\alpha, \gamma) + b(\beta, \gamma)$ (linearity property)
- 3) $(\alpha, \alpha) \geq 0$ & $(\alpha, \alpha) = 0 \Rightarrow \alpha = 0$ only (non-negative property)

where $\alpha, \beta, \gamma \in V$ & $a, b \in F$.

is known as "Inner product" on vector space $V(F)$.